

Refined Research Proposal

Learning Continuous Spectral Interaction Laws for Controllable Point Distributions

Research direction: retain the working neural-energy prototype, but refine the scientific contribution around continuous spectral conditioning, unseen-target generalization, learned interaction laws, spatial regularity, amortized efficiency, and progressive scalability.

1. Executive Summary

The current prototype learns a pairwise neural interaction energy conditioned on a target spectrum, obtains particle forces by differentiating that energy, evolves particles, rasterizes the final point set, computes its FFT/PSD, and trains through the spectral objective. This is an excellent foundation, but the basic mechanism alone should not be presented as the main novelty: neural interaction-energy and learned force-field constructions already have precedent.

The refined research question is stronger: **Can one conditional neural energy field learn a continuous family of particle interaction laws that generates point distributions for unseen target spectra, preserves measurable spatial regularity, exposes interpretable interaction behavior, and reduces the cost of producing new target distributions compared with optimizing every target from scratch?**

2. Research Problem

Point distributions are important in rendering, sampling, stippling, halftoning, and procedural graphics. Their spatial behavior can be characterized using power spectral density. Classical optimization can target a desired PSD, but normally requires a new optimization process for each target. The proposed method instead learns an interaction law once and conditions it on the desired spectral parameter γ .

The central distinction is therefore **amortized conditional generation**, rather than simply proving that a neural network can fit one spectrum.

3. Novelty Position

What is not sufficient as the primary novelty

- An MLP representing a pairwise energy.
- Computing force as the negative gradient of the energy.
- Backpropagating through unrolled particle dynamics.
- Optimizing a rasterized PSD loss alone.

What should become the research contribution

- **Continuous spectral interaction learning:** a single model represents a family of interaction laws across γ .
- **Unseen-spectrum generalization:** evaluate γ values not explicitly used during training.
- **Interaction-law interpretation:** analyze $E(r, \gamma)$ and $F(r, \gamma)$ rather than treating the network as a black box.
- **Learned regularity:** investigate whether appropriate short-range behavior emerges from training.
- **Amortized efficiency:** compare neural generation for new targets with classical per-target optimization.
- **Progressive generalization:** extend to local, anisotropic, and spatially varying spectra only after the core result is established.

4. Refined Pipeline

Target spectrum $\gamma \rightarrow$ condition encoder $\rightarrow$ conditional pairwise energy $E\theta(r_{ij}, \gamma) \rightarrow$ conservative force $F_i = -\partial E / \partial x_i \rightarrow$ unrolled dynamics $\rightarrow$ point set $X_T \rightarrow$ differentiable rasterization $\rightarrow$ FFT/PSD $\rightarrow$ spectral + regularity losses $\rightarrow$ backpropagation.

For the first version, $r_{ij} = ||x_i - x_j||$. The pairwise formulation is permutation invariant and makes the learned interaction law directly visualizable.

5. Conditional Neural Energy Model

Start with a compact MLP:

$$E_{ij} = \text{MLP}_{\theta}(r_{ij}, \gamma)$$

The total energy is summed over particle pairs and forces are obtained using automatic differentiation:

$$F_i = -\partial E_{\text{total}} / \partial x_i$$

- Start with a fully learned energy; do not assume a hand-designed repulsive potential.
- Use a small model so learned energy/force curves remain easy to inspect.
- Introduce a stabilizer only as a clearly reported fallback/ablation if collapse occurs.
- Track energy, force magnitude, minimum spacing, and particle displacement during dynamics.

6. Most Important Refinement: Continuous γ

Instead of training only on discrete blue/white/red-like categories, use γ as a continuous control variable. A practical starting range is $\gamma \in [-2, +2]$. Random γ sampling during training is preferable to only five fixed values.

- Train over a continuous/random distribution of γ .
- Hold out intermediate values such as -1.5 , -0.7 , 0.3 , 0.8 , and $+1.5$.
- Plot target γ against achieved spectral slope and PSD error.
- Measure whether spatial statistics change smoothly with γ .
- Test extrapolation separately from interpolation.

This experiment is the clearest way to show that the network learns a conditional interaction family rather than memorizing a small set of spectral classes.

7. Loss Function

Use a minimum combined objective:

$$L = \lambda_{\text{PSD}} L_{\text{log-PSD}} + \lambda_{\text{slope}} L_{\text{slope}} + \lambda_{\text{NND}} L_{\text{CV-NND}}$$

- **Log-PSD loss:** measures spectral fidelity while reducing domination by large spectral magnitudes.
- **Slope loss:** directly measures the target spectral trend.
- **NND coefficient of variation:** discourages pathological particle clustering.

The regularity term is important because differentiable Gaussian-splat rasterization can hide sub-kernel particle collapse from a purely image/PSD-based loss.

8. Learned Regularity Study

Run three controlled versions:

- A — spectral loss only.
- B — spectral + nearest-neighbor regularity.
- C — spectral + regularity + fixed hand-designed repulsion.

Compare PSD error, pair-correlation $g(r)$, NND CV, discrepancy/Ripley statistics, visual quality, and stability. If B learns sensible short-range behavior without fixed repulsion, that is stronger evidence that the energy field has learned useful interaction structure.

9. Interpret the Learned Energy

For a grid of γ values, produce:

- $E(r, \gamma)$ versus r .

- $F(r, \gamma)$ versus r .
- Heat maps of $E(r, \gamma)$ and $F(r, \gamma)$.
- Changes in attractive/repulsive regions as γ changes.

These figures should answer a central scientific question: *what interaction law did the network actually learn?*

10. Required Classical Baseline

Implement direct per-target classical particle-position optimization using the same domain, particle count, spectral objective, rasterization, FFT resolution, boundary handling, and regularity measures wherever possible.

- Measure final PSD/slope error.
- Measure spatial regularity.
- Measure optimization time for every new γ .
- Measure neural inference time for new γ after training.

Do not claim the neural approach is better simply because it works. The important comparison is quality, generalization, and amortized cost.

11. Dynamics and Stability

Begin with Euler or semi-implicit integration and approximately 50–100 unrolled steps. Monitor total energy, force norms, minimum spacing, loss, and displacement.

For ideal continuous gradient flow $dX/dt = -\nabla E$, $dE/dt = -\|\nabla E\|^2 \leq 0$. This does not by itself establish global convergence for the complete learned, discretized system, so the paper should report empirical stability rather than overclaim convergence.

12. Scalability

Full pairwise interaction is $O(N^2)$. Start with $N=256$ for validation. Then test larger point sets and introduce k -nearest-neighbor interactions with $k=16, 32$, and 64 .

- Compare runtime and quality.
- Test whether local interactions preserve the learned spectrum.
- Target $N=1,024$ and $N=4,096$ or larger when resources permit.

13. Advanced Extension: Anisotropic Spectra

After isotropic control is reliable, condition on $c=[\gamma, \alpha, \theta]$ and use the relative vector rather than distance alone:

$$E_{ij} = \text{MLP}_{\theta}(\Delta x_{ij}, c)$$

This allows non-radially-symmetric target PSDs and provides a stronger demonstration of conditional interaction learning.

14. Advanced Extension: Spatially Varying Spectra

Use a spatial condition $\gamma(x, y)$ so different regions of one domain request different spectral behavior. A compelling demonstration is a smooth red-like $\rightarrow$ white-like $\rightarrow$ blue-like transition in one point distribution. This should be attempted only after the global scalar- γ model is strong.

15. Evaluation Protocol

- **Spectral:** log-PSD MSE, slope error, frequency-band error.
- **Spatial:** pair-correlation $g(r)$, NND CV, discrepancy, and/or Ripley's K/L.
- **Generalization:** unseen interpolation and separate extrapolation tests.
- **Efficiency:** training cost, new-target generation time, classical per-target optimization time.
- **Robustness:** at least three random seeds, initialization changes, particle-count changes, and boundary variations.
- **Visual:** consistent point rendering plus at least one graphics/rendering application.

16. Essential Ablations

- Neural energy without stabilizer vs fixed repulsion.
- Energy-gradient force vs directly predicted force.
- Spectral-only vs spectral + regularity loss.
- Global pairwise vs kNN local interactions.
- Different boundary treatments.
- Discrete γ training vs continuous/random γ training.

17. Software Architecture

Module	Purpose
data.py	Target spectra and continuous γ sampling
energy.py	Conditional pairwise neural energy
dynamics.py	Force computation and particle integration
rasterize.py	Differentiable point rasterization
spectrum.py	FFT and PSD computation
losses.py	Spectral and spatial losses
baseline.py	Classical per-target optimizer
train.py	Training/checkpointing
evaluate.py	Metrics and ablations
visualize.py	PSD, point sets, energy/force plots

18. Initial Hyperparameters

- Domain: $[0,1]^2$.
- $N=256$ initially.
- γ range: $[-2,+2]$.
- 50–100 dynamics steps.
- Adam optimizer.
- Log-PSD + slope + NND regularity.
- At least three seeds.

Do not begin with Transformers, diffusion, 3D, or simultaneous anisotropic/spatial conditioning. The first objective is to establish a clean, reproducible two-dimensional core result.

19. Milestones

- **M1:** Validate differentiable rasterization, FFT and PSD gradients.
- **M2:** Train one target without instability.
- **M3:** Demonstrate measurable spectral control for $\gamma=+1$.
- **M4:** Demonstrate quantitative unseen- γ interpolation.
- **M5:** Complete classical baseline and quality/cost comparison.
- **M6:** Scale to $N \geq 4,096$ or the largest justified size using local interactions.
- **M7:** Demonstrate anisotropic control if the core is successful.
- **M8:** Demonstrate spatially varying control if results remain strong.
- **M9:** Demonstrate a graphics/rendering application.

20. Risk Management

- **Classical method is equally good:** emphasize continuous generalization, interaction interpretability, or amortized cost only if quantitatively supported.
- **Particle collapse:** inspect NND statistics; add regularity loss or a minimal stabilizer as an explicit ablation.
- **BPTT memory:** shorten unrolls, checkpoint, or move to kNN.
- **Rasterization artifacts:** validate gradients and vary kernel size/resolution.
- **Memorization:** use random γ training and strict held-out γ tests.
- **Uncertain novelty:** complete a targeted literature review before final publication claims.

21. Immediate 14-Day Plan

- **Days 1–2:** Validate differentiable rasterization, FFT and PSD loss.
- **Days 3–4:** Run $N=256$, $\gamma=+1$, purely learned pairwise energy, no stabilizer.
- **Days 5–6:** Inspect collapse, PSD convergence, spacing and E/F curves.
- **Days 7–8:** Train continuous/random γ and evaluate unseen γ .
- **Days 9–10:** Implement the classical per-target optimizer.
- **Days 11–14:** Compare neural/classical quality and cost; decide whether to proceed to kNN.

Decision gate: do not add anisotropy or spatial variation until unseen- γ generalization and the classical baseline are convincingly demonstrated.

22. Recommended Paper Narrative

- Different target spectra imply different desired spatial interaction behavior.
- A conditional neural energy can represent this behavior as a continuous interaction family.
- Unseen target spectra can be generated without training a separate model for each target.
- Energy/force plots reveal how the learned interaction law changes with spectral condition.
- Spatial statistics verify that spectral matching does not hide pathological clustering.
- A classical baseline establishes whether amortized learning is practically useful.
- Local interactions establish scalability.
- Anisotropic and spatially varying spectra provide progressive extensions.

23. Refined Research Claim

This work proposes learning a continuous family of particle interaction laws conditioned on a desired spatial spectrum. The objective is not merely to generate a few predefined spectral point sets, but to learn an interaction field that generalizes to unseen and intermediate targets, exhibits interpretable changes in energy and force with the spectral condition, preserves measurable spatial regularity, and amortizes the cost of generating new distributions. The method will be evaluated against direct classical per-target optimization and progressively extended to scalable local interactions, anisotropic spectra, and spatially varying spectral control.

24. Final Success Criterion

The project becomes a strong research contribution only if three results are simultaneously demonstrated: (1) quantitative unseen-spectrum generalization, (2) meaningful interpretation of the learned interaction law, and (3) a defensible quality/efficiency advantage or other clear benefit over classical per-target optimization. The current working prototype should therefore be retained as the foundation, while these experiments provide the refinement that turns it into a research study.